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**OFFLINE SIGNATURE VERIFICATION USING SIAMESE CONVOLUTIONAL
NEURAL NETWORKS**

**A Project Report Submitted in Partial Fulfillment of the Requirements for the Degree
of Bachelor of Science (B.Sc.) in Computer Science**

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ABSTRACT

Signature verification remains one of the most widely accepted and accessible methods for identity authentication in financial, legal, and administrative systems. However, manual verification is highly subjective and prone to human error, particularly when confronted with skilled forgeries. This project presents an automated Offline Signature Verification system based on a Siamese Convolutional Neural Network (CNN) architecture. Unlike traditional classification models, the Siamese network employs a twin-architecture with shared weights and a Contrastive Loss function (BCEWithLogitsLoss) to learn a robust 64-dimensional geometric feature space. This allows the system to compute the spatial similarity between a reference signature and a query signature, effectively identifying forgeries without requiring massive, user-specific training datasets.

The implementation features an automated preprocessing pipeline that utilizes Otsu's binarization and dynamic bounding-box normalization to eliminate background noise and standardize varied input sizes. Extensive data augmentation was applied to prevent overfitting and improve generalization across users. The model was trained over 29 epochs, with early stopping utilized to capture the optimal weights at a validation loss of 0.5605. During evaluation on the testing dataset, the system achieved an F1-Score of 75.18% and an Equal Error Rate (EER) of 28.57% at an optimized similarity threshold of 0.345. The results demonstrate that the Siamese architecture is highly effective at extracting unique biometric features from static images, providing a scalable and robust framework for automated offline signature authentication.

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF STUDY

Biometric authentication has become an indispensable component of modern security systems, providing reliable methods for verifying individual identity based on physical or behavioural characteristics (Veraros et al., 2026). Among the various biometric modalities, such as fingerprint scanning, facial recognition, and iris scanning, handwritten signatures hold a unique and historically significant position (Wang et al., 2026). For centuries, the handwritten signature has been the most socially and legally accepted method for authenticating documents, authorizing financial transactions, and validating administrative agreements (Adhikari et al., 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

Signature verification is generally classified into two categories: online (dynamic) and offline (static) verification (Chetry et al., 2025). Online signature verification utilizes specialized hardware, such as digital tablets and styluses, to capture dynamic features in real time, including pen pressure, stroke speed, and acceleration (Veraros et al., 2026). Conversely, offline signature verification deals exclusively with static images of signatures that have been scanned or photographed from a physical document (Chetry et al., 2025). Because it requires no specialized capturing hardware at the time of signing, offline verification is far more practical, cost-effective, and universally applicable, especially for legacy documents such as bank checks, legal contracts, and historical archives (Arrigoni, 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

However, offline signature verification is inherently more challenging than its online counterpart (Sukiman et al., 2025; Veraros et al., 2026). The absence of dynamic temporal data means that verification systems must rely entirely on the spatial geometry and structural topology of the ink strokes (Tehsin et al., 2026; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.). Furthermore, handwritten signatures naturally exhibit high intra-class variability, meaning a person's signature slightly changes every time they write it due to posture, mood, or writing surface, and low inter-class variability, where highly skilled forgeries closely mimic the original spatial structure (Xiao & Ding, 2022).

Traditionally, the task of verifying static signatures has been handled manually by forensic document examiners or bank tellers (Arrigoni, 2025). This manual approach is time-consuming, highly subjective, and prone to human error, particularly when operators suffer from fatigue or lack specialized forensic training (Halimi, 2025). As global financial transactions and digital document processing scale, the need for automated, objective, and high-speed offline verification systems has become critical (Sukiman et al., 2025).

In recent years, the rapid advancement of Artificial Intelligence (AI) and Deep Learning has provided powerful new tools for computer vision tasks (Sukiman et al., 2025). Convolutional Neural Networks (CNNs) have proven exceptionally capable of extracting complex hierarchical features from static images (Veraros et al., 2026; Wang et al., 2026). However, traditional CNN classifiers struggle in biometric authentication scenarios because they require large datasets of signatures for every single user to train effectively, which is practically impossible to obtain in real-world scenarios due to data scarcity ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.; Veraros et al., 2026). To solve this, researchers have turned to Siamese Neural Networks, a specialized architecture designed to learn the similarity between two inputs rather than classifying them into fixed categories (Adhikari et al., 2025; Dey et al., 2017). By utilizing contrastive learning, a Siamese network can effectively compare a newly presented query signature against a known reference signature, making it a highly robust and scalable solution for modern offline signature verification (Dey et al., 2017; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

1.2 PROBLEM STATEMENT

Despite the proliferation of digital security measures, paper-based documents and handwritten signatures remain the primary method of authorization in legal, administrative, and financial sectors (Adhikari et al., 2025). The manual verification of these static signatures by human clerks is fraught with challenges. It is a slow, subjective process that is highly susceptible to human error, fatigue, and inconsistency (Arrigoni, 2025; Halimi, 2025). When faced with skilled forgeries, where a forger meticulously copies the geometric structure of a signature, the human eye frequently fails to detect the fraud, leading to severe financial and legal consequences (Adhikari et al., 2025; Wang et al., 2026).

Automating this process using traditional machine learning algorithms has historically required exhausting manual feature engineering, such as calculating stroke angles, aspect ratios, and edge orientations, which struggles to adapt across different handwriting styles (Wang et al., 2026). While modern deep learning algorithms, specifically standard Convolutional Neural Networks (CNNs), have solved the manual feature extraction problem, they introduce a new critical flaw: they require thousands of training examples per individual class to function accurately (Veraros et al., 2026). In a real-world banking or legal scenario, it is practically impossible to collect thousands of signature samples from a single user (Arrigoni, 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

Therefore, there is a critical need for an automated, scalable system capable of learning the intrinsic spatial features of a signature from a very small number of reference samples, and effectively distinguishing those genuine signatures from highly skilled forgeries (Adhikari et al., 2025; Tehsin et al., 2026).

1.3 AIM OF THE STUDY

The primary aim of this study is to design, implement, and evaluate an automated Offline Signature Verification system using a Siamese Convolutional Neural Network (CNN) architecture capable of accurately distinguishing between genuine and forged signatures by comparing their geometric similarities in a lower-dimensional feature space.

1.4 OBJECTIVES OF THE STUDY

- ❖ To achieve the aforementioned aim, the following specific objectives will be pursued:
- ❖ To develop a robust image pre-processing pipeline (including grayscale conversion, Otsu's binarization, and dynamic bounding-box cropping) to eliminate background noise and standardize static signature images.
- ❖ To design a Siamese Convolutional Neural Network consisting of twin identical sub-networks capable of extracting deep spatial features from input signatures.
- ❖ To train the Siamese architecture using a contrastive loss function (Binary Cross-Entropy with Logits) on pairs of genuine and forged signatures, enabling the network to learn geometric similarities rather than fixed categorical classes.
- ❖ To implement data augmentation techniques that simulate natural intra-class variability, improving the model's ability to generalize to unseen users.
- ❖ To evaluate the performance, accuracy, and reliability of the proposed system using standard biometric metrics, including the Equal Error Rate (EER), F1-Score, and a Confusion Matrix analysis.

1.5 SCOPE OF THE STUDY

This research is strictly confined to offline (static) signature verification. It focuses exclusively on analyzing 2-dimensional scanned or photographed images of signatures and does not account for online (dynamic) properties such as writing speed, pen pressure, or stroke sequence.

The technical scope of the study encompasses the design and implementation of a custom Siamese Convolutional Neural Network (CNN) built using the PyTorch deep learning framework. The system is designed to evaluate signature pairs (a known reference and an unknown query) and output a similarity probability score. The evaluation of the system is constrained to a pre-collected dataset of genuine and forged signature pairs, and the performance is measured purely on its ability to classify these pairs correctly based on geometric feature extraction. The deployment of this model into a production-level enterprise application (such as a full banking web interface) falls outside the scope of this particular study.

1.6 SIGNIFICANCE OF THE STUDY

The findings and implementation of this study hold significant value for both the academic community and commercial industries. The key significances include:

Financial and Administrative Security: By automating the verification process, banks, clearing houses, and legal firms can drastically reduce the rate of fraudulent transactions and document tampering, saving millions in potential financial losses.

Reduction of Human Error: Implementing a mathematically objective neural network removes the subjectivity, fatigue, and inconsistency associated with human document examiners, leading to highly standardized and reliable security checks.

Advancement of Few-Shot Learning: Traditional AI systems require vast amounts of data per user, which is unfeasible in biometrics. This study demonstrates the practical viability of Siamese architectures for "few-shot learning," proving that AI can authenticate users securely using only one or two reference images.

Efficiency and Scalability: An automated offline verification system can process hundreds of scanned checks or documents per second. This study provides a foundational architecture that institutions can scale up to handle massive workloads far beyond human capability.

CHAPTER 2

LITERATURE REVIEW

2.1 SIGNATURE VERIFICATION

Signature verification is a behavioral biometric process concerned with determining whether a given handwritten signature belongs to the claimed individual or is a forgery produced by an impostor (Chetry et al., 2025; Xiao & Ding, 2022). It is widely applied in banking, legal documentation, contract authentication, and identity management systems ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks", n.d.; Veraros et al., 2026). Signature verification systems are broadly classified into two categories based on how the signature data is captured and processed: online (dynamic) verification and offline (static) verification (Chetry et al., 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks", n.d.)

Online Signature Verification

Online signature verification captures a signature as it is being written, typically using a digitizing tablet, stylus, smartphone touchscreen, or pen-input device (Veraros et al., 2026). Because the signature is captured in real time, the system has access to dynamic time-series features such as pen pressure, writing speed, stroke order, velocity, acceleration, and the time taken to complete the signature, in addition to the final shape of the signature itself ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.; Veraros et al., 2026). These temporal and kinematic features are difficult for a forger to reproduce even if they can visually copy the static shape of a signature, which generally makes online verification more accurate and more resistant to forgery than offline methods (Xiao & Ding, 2022). The original Siamese neural network architecture, in fact, was first proposed for exactly this task-verifying signatures written on a pen-input tablet (Bromley et al., 1993). The main drawback of online verification is that it requires specialized hardware at the point of signing, which limits its applicability to scenarios such as verifying signatures on paper documents, cheques, or historical records ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

Offline Signature Verification

Offline signature verification, by contrast, works with static, two-dimensional images of a signature, such as those scanned from a paper document, cheque, or contract ("State-of-the-Art -Offline Signature Verification Utilizing Siamese Neural Networks," n.d.; Veraros et al., 2026). No temporal or dynamic writing information is available; the system must rely entirely on visual and structural characteristics extracted from the image, including shape, stroke thickness, curvature, slant, and the overall pattern of the signature ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.; Veraros et al., 2026). This makes offline verification considerably more challenging than online verification because subtle but critical differences between a genuine signature and a skilled forgery must be captured from static pixels alone (Hafemann et al., 2017). Offline verification is, however, far more widely applicable in real-world settings, since most legal, financial, and administrative processes still rely on signatures affixed to physical or scanned documents ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.). Consequently, the majority of recent deep learning research, including work built around Siamese neural networks, has focused on improving the accuracy and generalization of offline signature verification systems (Dey et al., 2017; Jagtap et al., 2020).

2.2 SIGNATURE FORGERY

A central challenge in building a signature verification system is the variety of ways a signature can be forged. Forgeries are conventionally grouped into categories based on the forger's level of knowledge of, and access to, the genuine signature: random forgery, simple forgery, and skilled forgery (Chetry et al., 2025; Wang et al., 2026). The difficulty of detecting a forgery increases substantially as the forger's information about the genuine signature increases.

Random Forgery

A random forgery occurs when the forger has no knowledge of the genuine signature's shape or the signer's name, and simply uses their own natural signature, or an arbitrary signature, to impersonate another person (Chetry et al., 2025; Wang et al., 2026). Because the forged sample bears little to no resemblance to the genuine signature in terms of shape, size, or style and differs significantly in appearance, random forgeries are the easiest category for a

verification system to detect and are typically used to establish a baseline or best-case performance for a model (Chetry et al., 2025).

Simple Forgery

A simple forgery (sometimes called a casual forgery) is produced when the forger knows the name of the person being impersonated but has not seen or studied an example of that person's genuine signature (Chetry et al., 2025; Wang et al., 2026). The forger writes the target's name in their own natural handwriting style. Simple forgeries tend to show more resemblance to the genuine signature than random forgeries, since the correct name and letters are used, but the overall stroke pattern, rhythm, and style still differ noticeably from the genuine signer's, making detection moderately more difficult than random forgery but still comparatively tractable (Chetry et al., 2025)

Skilled Forgery

A skilled forgery is created when an imposter produces a highly deceptive signature after repeatedly observing or practicing the genuine signature multiple times (Chetry et al., 2025). Because the imitator has prior knowledge of the target style and replicates it closely, this is the most challenging category of forgery for offline signature verification systems to detect (Chetry et al., 2025; Wang et al., 2026)

2.3 MACHINE LEARNING

2.3 Machine Learning

Machine learning (ML) has historically played a foundational role in the development of offline signature verification systems. Traditional ML approaches predominantly relied on manually engineered, hand-crafted features to capture signature characteristics, utilizing geometric dimensions, texture descriptors, and gradient-based properties such as Histogram of Oriented Gradients (HOG) and Local Binary Patterns (LBP) (Adhikari et al., 2025; Chetry et al., 2025; Wang et al., 2026). These extracted features were subsequently fed into classical classification algorithms-including Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Random Forests-to differentiate between genuine and forged signatures (Adhikari et al., 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.). While these methodologies produced reasonable results under

controlled conditions, they were fundamentally constrained by their heavy dependence on manual feature engineering, which severely limited their adaptability across different users and datasets (Adhikari et al., 2025; Veraros et al., 2026).

Furthermore, they proved highly sensitive to acquisition noise, pen pressure variations, and writing inconsistencies, ultimately struggling to generalize across diverse signers or effectively detect subtle imitations like skilled forgeries (Adhikari et al., 2025; Wang et al., 2026). These inherent limitations motivated a necessary paradigm shift toward deep learning methodologies capable of automatic feature extraction (Chetry et al., 2025).

2.4 DEEP LEARNING

2.4 Deep Learning

Deep learning addresses the strict constraints of traditional machine learning by automatically extracting discriminative representations directly from raw signature images, entirely eliminating the need for explicit, manual feature design (Adhikari et al., 2025; Chetry et al., 2025). Within this subfield, Convolutional Neural Networks (CNNs) have emerged as the de facto standard for image-based tasks due to their capacity to model local structures and learn hierarchically abstract features (Veraros et al., 2026; Wang et al., 2026). This automatic, hierarchical feature learning has allowed deep learning systems to substantially outperform traditional machine learning approaches, particularly in writer-independent settings that require models to generalize to entirely new signers' unseen during training (Adhikari et al., 2025; Veraros et al., 2026). However, framing offline signature verification merely as a standard image recognition or fixed-category classification problem is fundamentally flawed, as it fails to fully capture the distinct writing styles and dynamic similarities required for authentication (Xiao & Ding, 2022). Signature verification inherently demands a similarity assessment between two input samples rather than rigid, multi-class categorization (Xiao & Ding, 2022). This structural limitation of standalone CNN classifiers serves as the primary motivation for adopting Siamese Neural Network (SNN) architectures, which excel at learning similarity metrics between pairs of samples by evaluating their Euclidean distance in a shared, embedded feature space (Adhikari et al., 2025; Dey et al., 2017).

2.5 SIAMESE NEURAL NETWORKS

A Siamese neural network is a specialized deep learning architecture designed for tasks that require comparing two inputs and determining how similar or dissimilar they are, rather than assigning an input to one of several fixed classes (Bromley et al., 1993; Xiao & Ding, 2022). This property makes Siamese networks particularly well suited to signature verification, where the essential question is always relational: does this signature match that reference signature closely enough to be considered genuine (Adhikari et al., 2025; Xiao & Ding, 2022)? Siamese networks are widely used in writer-independent verification systems because, once trained, they can compare signature pairs from signers who were never seen during training, without needing to retrain the model for every new user (Dey et al., 2017; Wang et al., 2026).

Architecture

The Siamese architecture consists of two or more identical sub-networks that share the same structure and, critically, the same set of weights and parameters (Bromley et al., 1993; Dey et al., 2017). Each sub-network independently processes one of the two input signature images and maps it into a fixed-length vector representation, often called an embedding, in a shared high-dimensional feature space (Dey et al., 2017; Xiao & Ding, 2022). Rather than producing a class label, the overall architecture outputs a distance or similarity score computed between the two embeddings, typically using a metric such as Euclidean distance (Adhikari et al., 2025; Jagtap et al., 2020). This score is then compared against a learned or fixed threshold to decide whether the input pair represents a genuine match or a forged mismatch (Dey et al., 2017; Xiao & Ding, 2022).

Twin Network

The two identical sub-networks in a Siamese architecture are commonly referred to as twin networks (Adhikari et al., 2025; Dey et al., 2017). Because the twins share identical weights, any change made to one network during training is mirrored exactly in the other, which guarantees that both networks encode any given input signature in a consistent way (Dey et al., 2017). This weight-sharing allows the network to learn a general notion of what makes two signatures similar or different rather than memorizing features specific to particular signers (Dey et al., 2017; Xiong & Cheng, 2021). Learning this relationship is essential for good generalization to unseen writers in writer-independent verification systems (Dey et al., 2017).

Feature Extraction

Within each twin network, convolutional layers extract discriminative visual features from the input signature image, such as stroke thickness, curvature, slant, spacing, and the overall pattern of pen movement, progressively transforming the raw pixel image into a compact embedding vector (Hafemann et al., 2017). Some recent architectures enhance this stage by incorporating additional scattering-based modules (Chokshi et al., 2023), or Transformer blocks (Ren et al., 2023), alongside standard convolutional layers. This strategic integration is designed to better capture both fine-grained local stroke details and the global structural layout of a signature simultaneously (Ren et al., 2023).

Similarity Learning

The Siamese network is trained using paired or sometimes tripled examples rather than individually labeled images (Dey et al., 2017). Common training objectives include contrastive loss, which pulls the embeddings of genuine positive pairs closer together while pushing genuine and forgery pairs further apart by minimizing and maximizing their Euclidean distances respectively (Dey et al., 2017). Another prominent objective is the triplet loss, which simultaneously considers an anchor signature, a positive genuine example, and a negative forged example, optimizing the embedding space so that genuine pairs are always closer than forged pairs by a predefined margin (Chattopadhyay et al., 2022). This similarity learning approach is what allows Siamese networks to perform one-shot or few-shot verification (Dey et al., 2017). As a result, the network does not need to be retrained for a new signer, since it has learned a general similarity function that can be applied to any new signature pair at inference time to reliably distinguish authentic signatures from forgeries (Dey et al., 2017).

2.6 RELATED WORKS

This section reviews prior research applying Siamese neural networks and related deep learning techniques to signature verification, summarizing for each study the method used, the dataset(s) evaluated, the accuracy achieved, and the key limitations identified. Full bibliographic details for each study cited below are provided in the References section at the end of this chapter.

Table 1: Comparative Analysis of Related Works

Citation	Method	Dataset	Accuracy	Limitation
Jagtap et al. (2020)	Siamese neural network (SNN) verifying genuine vs. forged offline signatures with enhanced embedding vectors.	GPDS Synthetic, MCYT-75, CEDAR	Improved performance over SigNet baseline on GPDS Synthetic (~86% range reported)	Trained on a reduced proportion of the original dataset size, raising questions about robustness at full scale.
Samsudin et al. (2024)	Two-stage Siamese network verifying original and enhanced signatures simultaneously; Focal Loss addresses genuine/forged class imbalance.	CEDAR, BHSig-Bengali, BHSig-Hindi	High accuracy reported across all three datasets, improving on standard Siamese baselines	Added architectural complexity (two stages) increases training cost; imbalance handling is dataset-specific and may need retuning for new corpora.
Xiong et al. (2022)	Siamese CNN-based twin-structure model computing Euclidean distance between genuine/forged embeddings.	GPDS, BHSig260-Hindi, BHSig260-Bengali	92% on GPDS synthetic (English); 80% Bengali; 78% Hindi – highest reported at the time versus prior Siamese baselines	Accuracy varies substantially by script/language, showing the model is sensitive to writing-system-specific stroke patterns.

Chattopadhyay et al. (2022)	Self-supervised, attention-guided reconstruction combined with dual triplet loss for writer-independent verification.	Standard offline benchmark datasets (CEDAR, GPDS, BHSig260)	Competitive results versus supervised Siamese baselines while reducing labelled-data requirements	Self-supervised pretraining adds pipeline complexity; attention mechanisms increase computational overhead.
Chokshi et al. (2023)	Combines Siamese architecture with scattering wavelet transforms for forgery detection and similarity assessment.	CEDAR and related offline benchmark sets	Reported competitive accuracy and improved robustness to noise versus plain CNN-Siamese models	Wavelet feature extraction adds preprocessing overhead; less end-to-end than pure deep-learning pipelines.
Ren et al. (2023); Majumder et al. (2023)	Two-channel, two-stream Transformer-based framework for offline signature verification, capturing both local and global stroke dependencies.	GPDS, CEDAR, BHSig260	Outperformed standard Siamese-CNN baselines on multiple benchmark datasets	Transformer components increase model size and training data requirements compared to lightweight CNN Siamese models.
ICCIT (2023–2024)	Hybrid Siamese-Transformer architecture with triplet loss; CNN extracts local features, Transformer models global dependencies; targets few-shot verification.	Public offline signature benchmark datasets (few-shot splits)	Superior performance reported in few-shot verification scenarios versus CNN-only Siamese baselines	Few-shot setting limits training data per writer, and Transformer components require careful regularization to avoid overfitting on small signature sets.

Wang et al. (2025)	Two-stage Siamese network incorporating a spatial transformer network to address positive/negative sample imbalance and improve spatial feature alignment.	Offline handwritten signature benchmark datasets	Improved feature learning and verification accuracy over conventional Siamese baselines, per Scientific Reports evaluation	Added spatial transformer stage increases architectural and computational complexity relative to a single-branch Siamese model.
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Taken together, these studies show a clear developmental trajectory in the field: from the original tablet-based Siamese network of Bromley et al. (1993), through convolutional Siamese networks such as SigNet (Dey et al., 2017) that established strong offline benchmarks on datasets like CEDAR and GPDS, to more recent hybrid architectures that combine Siamese networks with attention mechanisms (Majumder et al., 2023), wavelet scattering transforms (Chokshi et al., 2023), spatial transformers (Wang et al., 2025), and Transformer blocks (Ren et al., 2023) to better capture both local stroke detail and global signature structure. A recurring limitation across nearly all of the reviewed works is reduced generalization when models are evaluated across different datasets, scripts, or languages (Xiong et al., 2022), and continued difficulty in reliably distinguishing genuine signatures from skilled forgeries without access to large, balanced, and diverse training data (Hafemann et al., 2017). These gaps in cross-dataset generalization and skilled-forgery robustness represent the primary motivation for the approach adopted in this study.

CHAPTER 3

METHODOLOGY

3.1 INTRODUCTION

This chapter details the methodology adopted to develop a robust, deep learning-based signature verification system. Traditional image classification models are typically designed to categorize inputs into fixed, predefined classes, but offline signature verification fundamentally presents a different challenge (Xiao & Ding, 2022). The system must authenticate whether a given query signature matches a specific reference signature, often with very limited training samples per writer (Tehsin et al., 2026). To address this, the methodology relies on metric learning—specifically leveraging a Siamese Neural Network architecture (Wang et al., 2026).

A Siamese Network was chosen as the core of this system because it excels at one-shot and few-shot learning tasks (Adhikari et al., 2025). Instead of learning to classify a signature to a specific individual, which would require retraining the model every time a new user is added, the Siamese Network is trained to learn a similarity function (Tehsin et al., 2026). By taking two signatures as input simultaneously, the network learns to extract deep, meaningful feature embeddings and compute the absolute (L1) distance between them ("Siamese Network Methodology and Results for Signature Verification", n.d.). This approach allows the system to generalize to entirely new, unseen writers with high accuracy, making it highly scalable and robust against skilled forgeries (Adhikari et al., 2025).

The overall workflow of the proposed system is systematically broken down into several critical phases. It begins with comprehensive data collection, followed by a data pre-processing pipeline that includes cropping, grayscale conversion, resizing, and thresholding to standardize the input ("Siamese Network Methodology and Results for Signature Verification", n.d.; Wang et al., 2026). Subsequently, pair generation is employed to create positive (genuine-genuine) and negative (genuine-forged) signature pairs necessary for Siamese learning (Adhikari et al., 2025). The model training phase utilizes these pairs to optimize the network's feature extraction and distance computation capabilities (Dey et al., 2017). Finally, the evaluation and signature verification phase determine the system's efficacy by analysing probability scores against an optimized Equal Error Rate (EER) decision threshold to output a final "genuine" or "forged" prediction ("Siamese Network Methodology and Results for Signature Verification", n.d.; Chetry et al., 2025).

3.2 SYSTEM OVERVIEW

The signature verification system is designed as an end-to-end metric learning pipeline driven by a Siamese Neural Network. Rather than relying on manual, handcrafted feature engineering, the system autonomously learns to identify the highly distinct, writer-specific

stroke characteristics of a signature. The complete workflow of the system is logically structured into six sequential phases:

i. Data Collection: Raw signatures, encompassing both genuine samples and intentionally simulated forgeries, are gathered and digitized. This creates the foundational dataset required for the network to learn both intra-class variations (the natural, day-to-day variations in a single person's genuine signature) and inter-class similarities (the specific ways forgers attempt to mimic the reference signatures).

ii. Data Pre-processing: To ensure the Siamese model focuses purely on the structural strokes of the signature rather than background noise, artifacts, or inconsistent lighting conditions, the raw images undergo a strict preprocessing pipeline. This includes converting the images to grayscale, applying Otsu's thresholding to cleanly isolate the pen strokes into a pure binary (black-and-white) format, and resizing the images to standardized dimensions to be consumed by the network's convolutional layers.

iii. Pair Generation: Because the Siamese Network requires two simultaneous inputs to learn a similarity metric, the preprocessed images are dynamically structured into pairs during training. The system systematically generates "Positive Pairs" (consisting of two genuine signatures from the same writer) and "Negative Pairs" (consisting of a genuine reference signature paired with a forged attempt).

iv. Siamese Model Training: The generated image pairs are fed into twin, weight-sharing Convolutional Neural Network (CNN) branches. Each branch acts as a feature extractor, compressing the signature into a dense, low-dimensional embedding vector. The network computes the L1 (absolute) distance between these two embedding vectors and passes the resulting distance array into a final, fully connected classifier layer. The entire architecture is optimized end-to-end using Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss).

v. Model Evaluation: The trained network's generalization capability is rigorously assessed against unseen validation and test datasets. The model is evaluated across standard classification metrics including Accuracy, Precision, Recall, and F1-Score. More importantly, the evaluation process sweeps across a spectrum of decision thresholds to determine the Equal Error Rate (EER) - identifying the optimal threshold where the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are balanced.

vi. Signature Verification (Inference): During a real-world, production scenario, the system is provided with a known reference signature and a new, unverified query signature. The model extracts their features, computes their L1 embedding distance, and applies a sigmoid activation to output a final match probability. If this probability exceeds the optimized decision threshold identified during evaluation (e.g., 0.345), the system verifies the signature as a genuine match; otherwise, it is immediately flagged as a forgery.

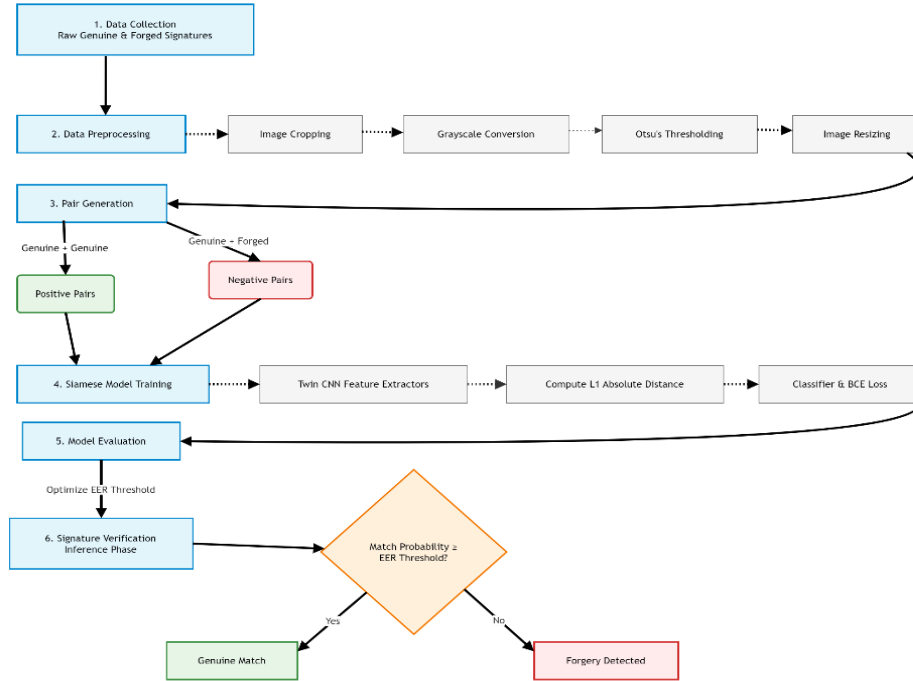


Figure 1: System Overview

3.3 DATASET DESCRIPTION

3.3.1 DATA SOURCE

The system utilizes an offline signature dataset structured specifically to support metric learning for forgery detection. The dataset collection procedure was designed to capture both the natural variability of an individual's signature and the deliberate attempts made by others to copy them.

The dataset consists of signatures collected from 105 different participants (writers). During the collection procedure, each participant provided multiple genuine signatures to establish a strong reference profile. Subsequently, skilled forgeries were collected by having other individuals attempt to replicate the genuine signatures. For every writer in the dataset, exactly 10 genuine signatures and 5 forged signatures were collected, ensuring a consistent volume of data across all participant profiles.

3.3.2 DATASET STRUCTURE

To facilitate seamless pair generation and data loading during the training pipeline, the dataset is meticulously organized into a hierarchical folder structure. The root directory contains 105 subdirectories, one corresponding to each unique writer (e.g., SIG_0001 through SIG_0105).

Within each writer's specific folder, the images are explicitly separated into genuine and forged sub-folders. This distinct separation is crucial for correctly labelling the positive and negative pairs during the Siamese network's training phase.

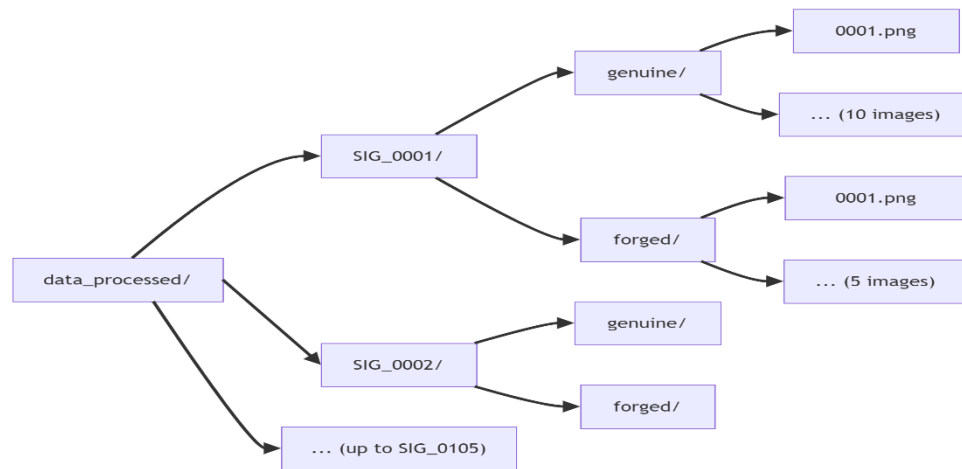


Figure 2: Dataset Folder Structure

3.3.3 DATASET STATISTICS

In total, the dataset contains 1,575 standardized signature images. All images undergo a pre-processing pipeline that resizes them to a fixed target resolution of 220×155 pixels (Width $\times$ Height) while preserving their original aspect ratio through padding. This ensures that the Siamese Convolutional Neural Network receives uniform input tensors without distorting the structural integrity of the signatures.

A comprehensive summary of the dataset statistics is provided in the table below:

Table 2: Summary of Dataset Statistics

Parameter	Quantity / Specification
Number of Writers	105
Genuine Signatures per Writer	10
Forged Signatures per Writer	5
Total Genuine Signatures	1,050
Total Forged Signatures	525
Total Image Count	1,575
Image Format	PNG / JPG
Processed Image Resolution	220×155 pixels (W $\times$ H)

3.4 DATA PRE-PROCESSING

Raw signature scans natively contain substantial background noise, varying resolutions, and inconsistent lighting, which can severely distort the spatial feature extraction process of convolutional layers. To ensure the Siamese Network focuses purely on the structural geometry of the signature strokes, a strict, mathematically grounded pre-processing pipeline was applied to all images prior to network ingestion.

3.4.1 IMAGE CROPPING

Raw scans often feature vast, uninformative white borders surrounding the signature. To maximize the computational efficiency of the CNN and focus entirely on stroke morphology, an automated bounding-box cropping algorithm is applied.

Let the binary image matrix be $I_B(x, y)$, where ink strokes are represented by 0 (black) and background by 255 (white). The bounding coordinates of the signature region are defined as:

$$x_{min} = \min_x | I_B(x, y) = 0 \text{ and } x_{max} = \max_x | I_B(x, y) = 0$$
$$y_{min} = \min_y | I_B(x, y) = 0 \text{ and } y_{max} = \max_y | I_B(x, y) = 0$$

The image is cropped to this region. To prevent the convolutional filters from encountering destructive boundary edge-effects, a fixed spatial padding $p = 10$ pixels is applied symmetrically around the cropped region, bounded by the original image dimensions.

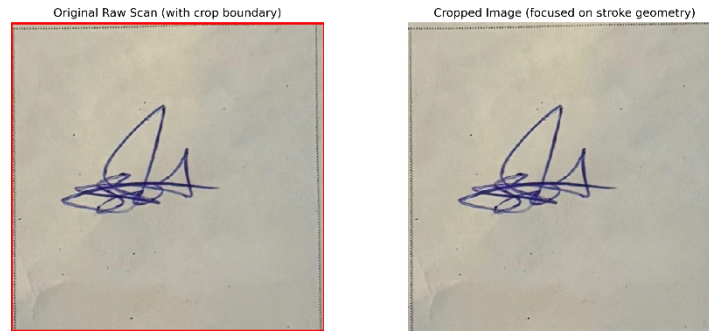


Figure 3: Original Image vs Cropped Image

3.4.2 Grayscale Conversion

Because the structural identity of a signature is defined by stroke geometry rather than ink color, the RGB channels provide redundant information. The $H \times W \times 3$ RGB tensor is projected into a $H \times W \times 1$ grayscale matrix I_{gray} , drastically reducing computational and memory overhead (Veraros et al., 2026; Xiao & Ding, 2022). The transformation is computed

using the standard luminance weighting:

$$I_{gray}(x, y) = 0.299 \cdot R(x, y) + 0.587 \cdot G(x, y) + 0.114 \cdot B(x, y)$$

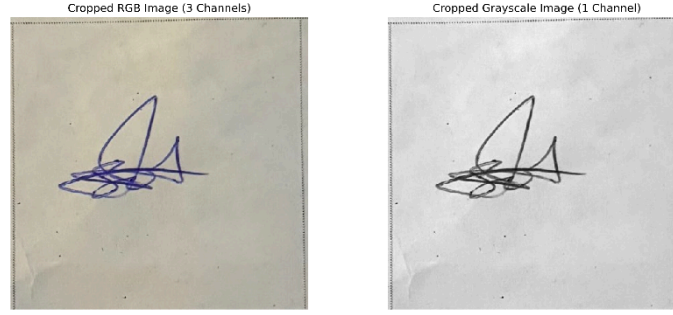


Figure 4: RGB vs Grayscale Image

3.4.3 Image Binarization (Otsu's Method)

To strictly standardize the contrast regardless of the scanner or pen used, the grayscale image I_{gray} is converted into a pure binary format (Shih et al., 2024; Wang et al., 2026). Instead of relying on a hardcoded threshold, Otsu's Thresholding algorithm is utilized to compute an optimal, image-specific threshold τ^* (Otsu, 1979; Xiao & Ding, 2022).

Otsu's method achieves this by searching exhaustively for the threshold τ that minimizes the intra-class variance $\sigma_w^2(\tau)$ of the foreground (ink) and background (paper) pixels (Otsu, 1979). Once τ^* is found, the binary mapping is applied as (Otsu, 1979):

$$I_B(x, y) = 0, \text{ if } I_{gray}(x, y) \leq \tau^* \text{ (Ink)}; 255, \text{ if } I_{gray}(x, y) > \tau^* \text{ (Background)}$$

Furthermore, an automated morphological check ensures the background remains strictly white (255) (Xiao & Ding, 2022). If the summation of black pixels exceeds white pixels (indicating a color-inverted scan), a bitwise inversion $I'_B = 255 - I_B$ is systematically applied to standardize the input (Wang et al., 2026).

3.4.4 Image Resizing

Convolutional Neural Networks require input tensors of uniform spatial dimensions. However, directly resizing signatures of arbitrary sizes $h \times w$ into the network's target dimensions ($H_{target} = W_{target} = 22$) would cause severe geometric distortion, altering the signature's unique aspect ratio.

To preserve the stroke geometry, an aspect-ratio-preserving padding strategy is applied. An isotropic scaling factor S is computed to fit the largest dimension: $S = \left(\frac{H_{target}}{h}, \frac{W_{target}}{w} \right)$. The signature is resized using area interpolation to new dimensions

$h' = \lfloor h \cdot S \rfloor$ and $w' = \lfloor w \cdot S \rfloor$. This resized matrix is then perfectly centered onto a uniform white canvas $C \in 255^{155 \times 220}$, utilizing spatial offsets

$$x_{offset} = \lfloor \frac{W_{target} - w}{2} \rfloor \text{ and } y_{offset} = \lfloor \frac{H_{target} - h}{2} \rfloor.$$

3.4.5 Image Normalization

To ensure smooth convergence during gradient descent, the pixel magnitudes must be scaled to prevent massive, unstable layer activations. The padded image C is transformed into a normalized float32 tensor I_{norm} , bounded strictly within the range $[0.0, 1.0]$:

$I_{norm}(x, y) = \frac{C(x, y)}{255.0}$. This unit-scaled tensor serves as the final, sanitized input delivered to the Siamese Network's convolutional layers.

3.4.6 Pair Generation

Because the Siamese architecture is explicitly designed for metric similarity learning (Bromley et al., 1993; Koch et al., 2015), it requires simultaneous twin inputs. This necessitates an exhaustive Pair Generation protocol, transforming the standardized dataset into a combinatorial pair-wise format (X_A, X_B, Y) , where Y represents the ground-truth relationship (Adhikari et al., 2025).

- **Positive Pair** ($Y = 1$): A reference signature X_A paired with a distinct genuine signature X_B from the same writer. The network is penalized if the computed embedding distance is large (Adhikari et al., 2025).
- **Negative Pair** ($Y = 0$): A reference signature X_A paired with a forged signature X_B . The network is penalized if the computed embedding distance is small (Adhikari et al., 2025).

Fundamental Role in Metric Learning: Pair generation shifts the neural objective from rigid multi-class categorization to dynamic similarity assessment (Xiao & Ding, 2022). By forcing the network to minimize the L1 embedding distance for $Y = 1$ pairs and maximize it for $Y = 0$ pairs, the Siamese Network learns the nuanced, writer-specific structural discrepancies (such as stroke velocity and curve pressure) distinguishing genuine signatures from highly skilled forgeries (Adhikari et al., 2025).

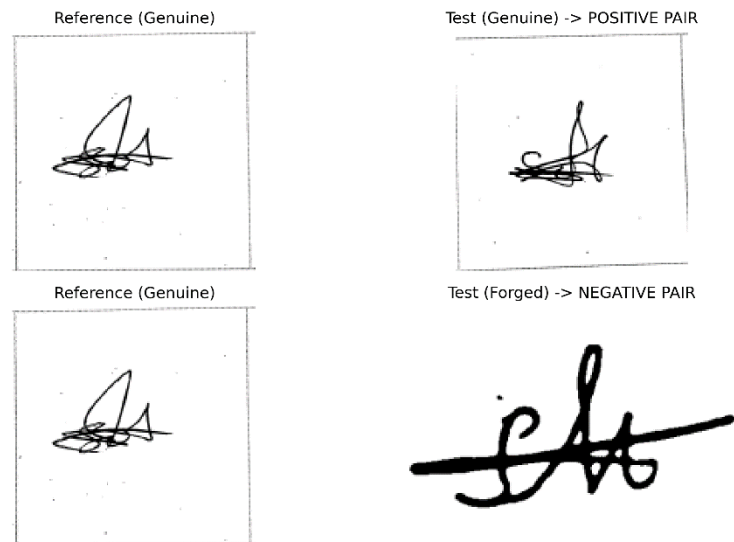


Figure 5: Pair Generation Process (Insert pair examples diagram here)

3.5 SIAMESE NETWORK ARCHITECTURE

To overcome the limitations of traditional image classification, this project implements a Siamese Neural Network (Xiao & Ding, 2022). A traditional neural network is designed to classify an image into a predefined category, but signature verification fundamentally presents a different challenge: the system must authenticate whether a new query signature matches a specific user, even if the network has never seen that user during training (Adhikari et al., 2025; "State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.).

A Siamese Network solves this by acting as a metric learning model rather than a traditional classifier ("State-of-the-Art Offline Signature Verification Utilizing Siamese Neural Networks," n.d.; Wang et al., 2026). It contains two identical "twin" subnetworks that share the exact same configuration, weights, and parameters (Dey et al., 2017; Halimi, 2025). By passing a reference signature into one twin and a query signature into the other, the network learns to extract their core structural features and mathematically compute how similar they are by projecting them into a shared embedding space (Dey et al., 2017; Xiao & Ding, 2022).

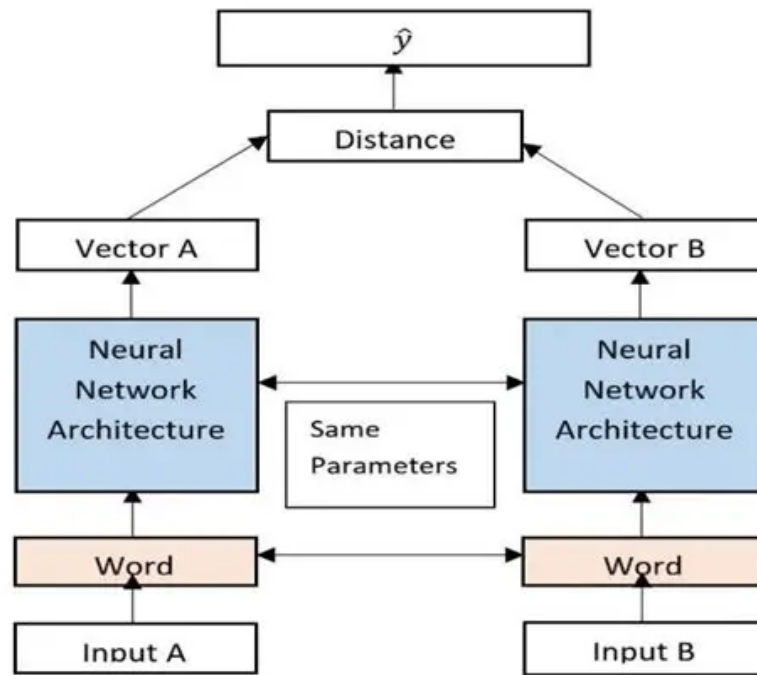


Figure 6: Siamese Network Architecture (Insert Siamese Architecture Diagram here)

3.5.1 Feature Extraction

The foundation of the Siamese Network is the shared Convolutional Neural Network (CNN) encoder (Xiao & Ding, 2022). Instead of memorizing a signature, this encoder is designed to extract deep geometric patterns (like stroke curvature and line intersections) from the 155×220 signature images (Adhikari et al., 2025; Xiao & Ding, 2022).

The feature extraction process flows through several stages:

- 1. Convolutional Backbone:** The image passes through sequential convolutional blocks. Each block applies filters to detect increasingly complex patterns (Xiao & Ding, 2022). To prevent the model from simply memorizing the training data (overfitting), Max Pooling layers are used to compress the image size, and Dropout layers randomly turn off a percentage of neurons during training (Adhikari et al., 2025; Xiao & Ding, 2022).
- 2. Global Average Pooling:** Once the spatial features are extracted, an Adaptive Average Pooling layer compresses the data, reducing the massive feature maps into a dense, manageable flat array (Adhikari et al., 2025; Xiao & Ding, 2022).
- 3. Embedding Generation:** The flattened array is passed through fully connected linear layers, which ultimately condense the signature's identity down to a highly concentrated 64-dimensional embedding vector (Adhikari et al., 2025; Xiao & Ding, 2022).
- 4. L2 Normalization:** Finally, this 64-dimensional embedding is L2-normalized. This mathematical scaling ensures that all extracted embeddings sit on the same scale (a "unit sphere"), making it much easier and more stable for the network to compare them later using distance metrics (Adhikari et al., 2025; Xiao & Ding, 2022).

3.5.2 Distance Computation

Once the twin networks have generated the 64-dimensional embeddings for Signature A (E_A) and Signature B (E_B), the network must compute the difference between them.

Instead of outputting a single number representing the overall distance, the model computes an element-wise *L1 (Absolute) Distance*. This means the network compares the two embeddings dimension-by-dimension, creating a new 64-dimensional vector that represents exactly where and how much the two signatures differ ("Siamese Network Methodology and Results for Signature Verification", n.d.).

The mathematical expression for the L1 absolute distance vector D is: $D = |E_A - E_B|$ More specifically, for each of the 64 features (where $i = 1, 2, \dots, 64$): $D_i = |E_{A,i} - E_{B,i}|$

Why L1 Distance? By preserving the distance as a full vector rather than a single summed number, the model allows the final classifier to learn which specific feature differences are acceptable (natural human variation) and which differences are strong indicators of a forgery (Koch et al., 2015).

3.5.3 Output Layer

After computing the L1 distance vector between the two signatures, the data is passed into a final fully connected output block. This block acts as the system's "judge," evaluating the distance vector to produce a final, raw output value known as a logit (z).

During Model Training: The raw logit z is fed directly into the [BCEWithLogitsLoss](#) function (Binary Cross-Entropy). This loss function is highly optimized and numerically stable for training. It evaluates how far off the model's prediction was from the ground-truth label (1 for a genuine match, 0 for a forgery) and updating the network's weights accordingly (Xiao & Ding, 2022).

During Inference (Real-World Verification): When the model is actually used to verify a signature, the raw logit z isn't very user-friendly. Therefore, it is explicitly passed through a Sigmoid activation function to convert it into a highly readable probability P , ranging between 0 and 1: $P = \sigma(z) = \frac{1}{1+e^{-z}}$ This probability P represents the network's confidence that the two signatures belong to the exact same writer. Finally, the system compares P against a pre-calibrated Equal Error Rate decision threshold (τ). The final verification rule is simple:

$$\text{Prediction} = \{\text{Genuine if } P \geq \tau, \text{ Forged if } P < \tau\}$$

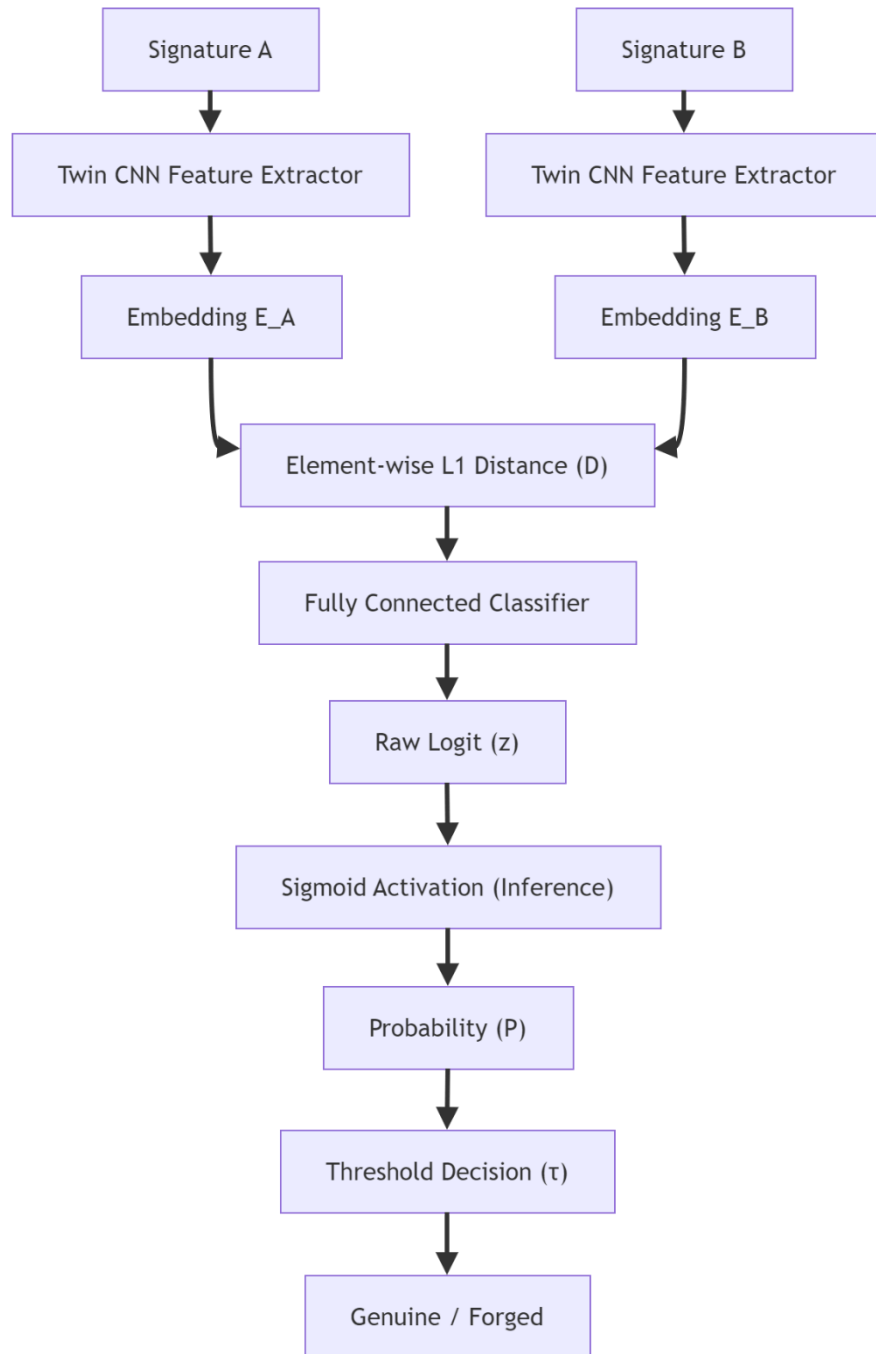


Figure 7: Inference Workflow Diagram:

3.6 MODEL TRAINING

Training a Siamese Network requires a carefully orchestrated configuration to ensure the model learns a robust, generalized similarity metric without memorizing the training data. This section details the exact datasets, hyperparameters, and optimization strategies used during the actual model training phase on a CUDA-enabled GPU.

3.6.1 Dataset Splitting and Pair Generation

To ensure the model is evaluated on completely unseen writers, the 105 participants in the dataset were strictly split at the subject level. During the training run, the exact data distribution was as follows:

Training Dataset (84 Subjects): The training dataset dynamically generated 7,980 signature pairs. Of these, 3,780 were positive pairs (genuine-genuine) and 4,200 were negative pairs (genuine-forged). Data augmentation was heavily applied during training to introduce variations (e.g., slight rotations or shifts), vastly increasing the robustness of the model.

Validation Dataset (10 Subjects): Used strictly to monitor the model's generalization capability during the training loop. This set consisted of 950 signature pairs (450 positive, 500 negative). Data augmentation was completely disabled here to test the model on pure, unaltered data.

Test Dataset (11 Subjects): A completely isolated holdout set containing the remaining subjects, reserved exclusively for final evaluation and Equal Error Rate (EER) threshold optimization after training.

3.6.2 Optimization and Loss Strategy

Optimizer: The model was optimized using AdamW (Adam with Weight Decay). A weight decay of 10^{-5} was applied to prevent the model from assigning excessively large weights to specific features (Chetry et al., 2025).

Loss Function: The network utilized the Binary Cross-Entropy with Logits Loss (BCEWithLogitsLoss) ("Siamese Network Methodology and Results for Signature Verification", n.d.). Crucially, because there was a slight class imbalance (4,200 forged pairs vs 3,780 genuine pairs), a dynamic pos_weight of 0.9000 was calculated and applied directly to the loss function. This mathematical weighting ensures the network pays equal attention to both classes, preventing the model from becoming biased toward the majority class (Xiao & Ding, 2022).

Gradient Clipping: To maintain stability during backpropagation, gradient clipping was enforced with a maximum norm of 1.0 (Pascanu et al., 2013).

3.6.3 Learning Rate Scheduling & Early Stopping

The network began with an initial learning rate of 3×10^{-4} . A ReduceLROnPlateau scheduler was employed to automatically cut the learning rate in half (factor of 0.5) if the validation loss failed to improve for 3 consecutive epochs (Halimi, 2025). To prevent overfitting, an Early Stopping mechanism was implemented with a patience of 10 epochs.

Actual Run Results: While the model was configured to train for a maximum of 50 epochs, the Early Stopping mechanism successfully halted training at Epoch 29. The absolute lowest validation loss (0.5605) was achieved at Epoch 19, and those specific weights were automatically saved and restored as the final production model.

3.6.4 Training Configuration Summary

The complete set of hyperparameters and actual execution metrics from the Siamese Network training run are summarized in the table below:

Table 3: Model Training Parameters

Parameter	Value / Configuration
Hardware Accelerator	CUDA GPU (Google Colab)
Maximum Epochs configured	50
Actual Epochs executed	29 (Stopped due to Early Stopping)
Best Validation Loss	0.5605 (Achieved at Epoch 19)
Batch Size	16 (499 Training Batches, 60 Validation Batches)
Initial Learning Rate	3×10^{-4}
Optimizer	AdamW (Weight Decay: 10^{-5})
Loss Function	BCEWithLogitsLoss (pos_weight = 0.9000)
Learning Rate Scheduler	ReduceLROnPlateau (Factor: 0.5, Patience: 3)
Early Stopping Patience	10 Epochs
Data Augmentation	Enabled (Training set only)

3.7 SOFTWARE AND HARDWARE REQUIREMENTS

The successful implementation, training, and evaluation of the Siamese Network required a specific stack of software tools and hardware resources. The development was split between local environments for scripting and cloud environments for heavy computational processing.

3.7.1 Software Requirements

Python (3.8+): The core programming language used for the entire project due to its extensive ecosystem for machine learning and computer vision.

PyTorch & Torchvision: The primary deep learning frameworks utilized to build, train, and evaluate the Siamese Network. PyTorch was chosen for its dynamic computation graph and native support for custom loss functions like BCEWithLogitsLoss.

OpenCV & NumPy: Utilized heavily in the preprocessing pipeline (preprocess.py) for highly optimized matrix operations, image resizing, and Otsu's thresholding binarization.

Scikit-Learn: Employed during the evaluation phase to calculate classification metrics and compute the Equal Error Rate (EER) threshold via the ROC curve.

Google Colab: A cloud-based Jupyter Notebook environment utilized specifically to leverage free, high-performance GPUs for training the model over massive datasets without local hardware constraints.

Visual Studio Code (VS Code): The primary Integrated Development Environment (IDE) used for local code authoring, debugging, and building the inference scripts.

Git: Utilized for version control to track codebase changes and maintain repository organization.

3.7.2 Hardware Requirements

Training a deep Convolutional Neural Network on thousands of image pairs requires significant computational power. The following hardware specifications were utilized (primarily provisioned via Google Colab):

Graphics Processing Unit (GPU): An NVIDIA GPU (typically a Tesla T4 provided by Google Colab) was strictly required for the training phase. The GPU accelerated the massive tensor multiplications and backpropagation steps, drastically reducing training time from days down to hours or minutes.

Central Processing Unit (CPU): A multi-core CPU was necessary to handle the heavy data-loading bottlenecks. The CPU managed the real-time data augmentation and dynamic pair generation (PairDataset) in parallel before passing the batches to the GPU.

Random Access Memory (RAM): A minimum of 12 GB of system memory was required to load the training batches, manage the validation data structures, and handle the memory-intensive image preprocessing pipeline without encountering out-of-memory (OOM) errors.

3.8 Chapter Summary

This chapter detailed the comprehensive methodology implemented to develop a highly robust, deep learning-based signature verification system. The system shifts away from traditional multi-class image classification and instead utilizes a metric-learning approach powered by a Siamese Neural Network.

The methodology began by outlining the data collection process, which sourced 1,575 genuine and forged signatures across 105 unique writers. To ensure the model focused strictly

on stroke geometry rather than background noise, a rigorous preprocessing pipeline was established. This pipeline included bounding-box cropping, grayscale conversion, Otsu's thresholding for optimal binarization, and aspect-ratio-preserving resizing.

The core of the system, the Siamese Network, was detailed extensively. By feeding dynamically generated positive and negative image pairs into twin Convolutional Neural Network (CNN) branches, the model was trained to extract deep spatial features into a normalized 64-dimensional embedding. The absolute (L1) distance between these embeddings was then computed and passed through a Fully Connected classifier to output a match probability.

Finally, the chapter outlined the specific training configurations-including the use of the BCEWithLogitsLoss function, AdamW optimization, dynamic learning rate scheduling, and early stopping on a CUDA-enabled GPU-ensuring the model successfully generalized to unseen signatures without overfitting.

With the architectural foundation and training pipeline fully established, the next chapter will present the performance outcomes, evaluation metrics, and practical verification results achieved by this system.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 INTRODUCTION

Following the successful design and training of the Siamese Neural Network, this chapter presents a comprehensive evaluation of the system's performance. The primary objective of this evaluation is to determine how effectively the model can distinguish between genuine signatures and skilled forgeries on completely unseen writers.

The evaluation process is divided into two main phases. First, the training and validation phase is analyzed to assess the model's learning behaviour, convergence stability, and resistance to overfitting. Second, the finalized model is subjected to a rigorous evaluation using standard classification metrics, including Accuracy, Precision, Recall, and the F1-Score. Finally, to ensure the system meets biometric verification standards, an Equal Error Rate (EER) analysis is conducted to pinpoint the optimal decision threshold for real-world deployment.

4.2 TRAINING RESULTS

The Siamese Network was trained over a heavily augmented dataset of 7,980 training pairs, while its generalization was continually monitored against an unaltered validation dataset of 950 pairs. The training process utilized an Early Stopping mechanism configured with a patience of 10 epochs to automatically halt training when the validation loss ceased to improve, ensuring the model did not overfit.

The training officially halted at Epoch 29. However, the absolute best performing weights representing the lowest validation loss were achieved at Epoch 19 and were subsequently saved as the final production model.

The key performance metrics recorded at this optimal checkpoint (Epoch 19) were:

- ❖ Training Loss: 0.6278
- ❖ Training Accuracy: 61.45%
- ❖ Validation Loss: 0.5605
- ❖ Validation Accuracy: 72.63%

4.2.1 Observations from the Training Curves

The progression of the loss and accuracy metrics throughout the 29 epochs reveals several critical insights into the model's learning behaviour:

Stable Convergence: Both the training and validation loss steadily decreased during the first 15 epochs. The validation loss dropped rapidly from an initial 0.6560 down to 0.5605, indicating that the BCEWithLogitsLoss function was successfully minimizing the L1 distance for genuine pairs while maximizing it for forged pairs.

Generalization: Interestingly, the validation accuracy consistently outperformed the training accuracy (e.g., 72.63% vs 61.45% at Epoch 19). This is a direct, positive result of the heavy data augmentation applied exclusively to the training set. Because the training pairs were actively rotated, shifted, and altered to intentionally confuse the network, the training accuracy was artificially suppressed. When the network was evaluated on the clean, unaugmented validation pairs, it demonstrated a strong capability to generalize.

Early Stopping Activation: After Epoch 19, the validation loss began to slightly plateau and marginally fluctuate (ranging between 0.5607 and 0.5719). After 10 consecutive epochs without surpassing the baseline set at Epoch 19, the early stopping mechanism correctly terminated the run, preventing the model from beginning to memorize the training data.

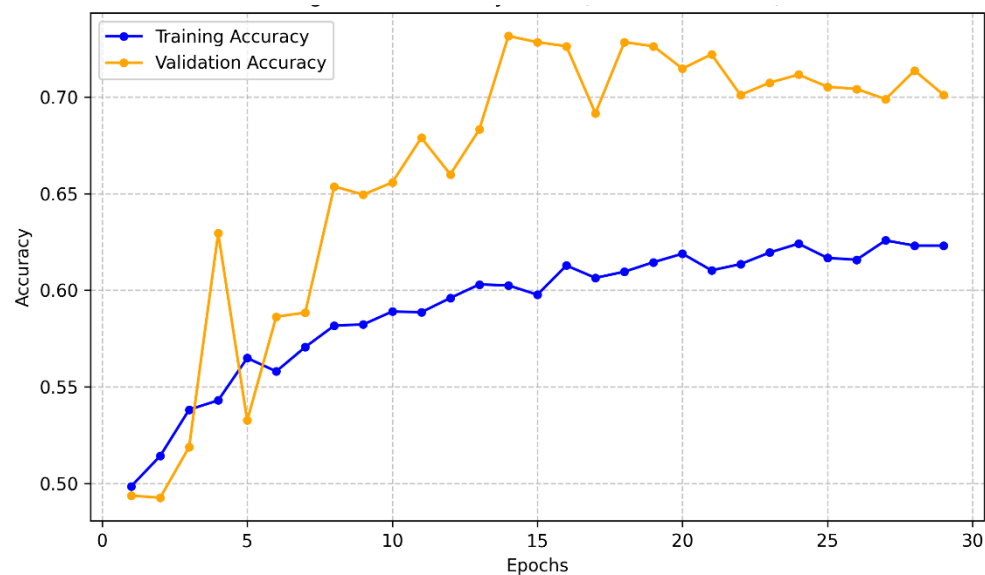


Figure 8: Accuracy Curve

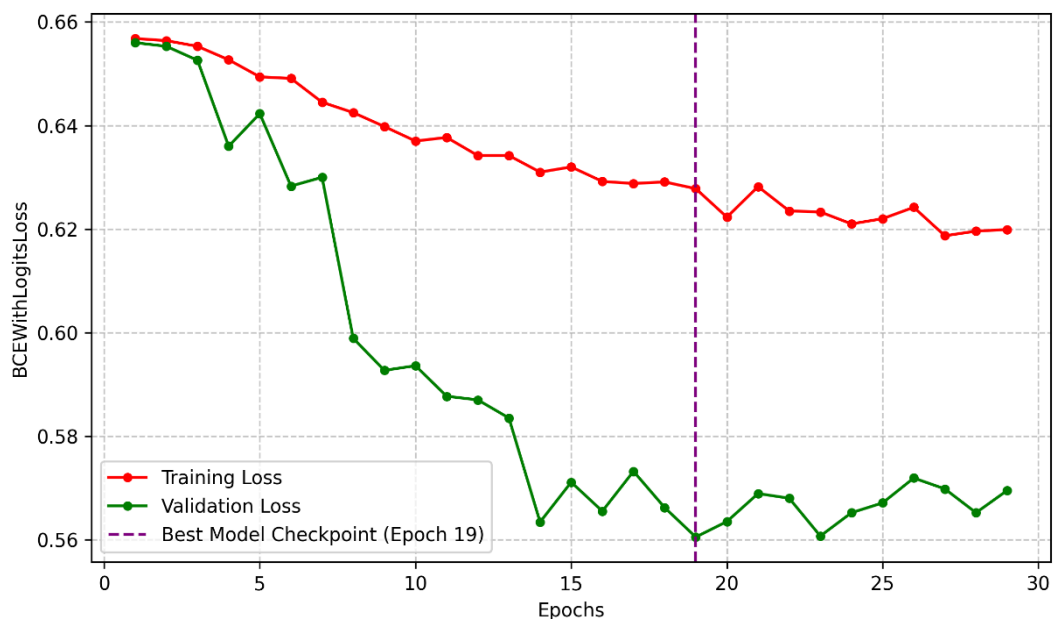


Figure 9: Loss Curve

4.3 MODEL EVALUATION

To rigorously test the model's ability to verify signatures in a real-world scenario, the fully trained Siamese Network was evaluated against an isolated Test Dataset. This holdout set consisted of 1,045 pairs (495 genuine, 550 forged) generated from 11 entirely unseen writers.

Because the Siamese Network outputs a continuous probability rather than a hard class label, evaluating its performance heavily depends on the chosen decision boundary.

4.3.1 Threshold Selection

Unlike traditional multi-class classification where a network simply outputs the highest-scoring class, a Siamese metric-learning model outputs a continuous similarity probability $P \in [0, 1]$. To classify a pair as "Genuine" or "Forged", a specific decision threshold (τ) must be mathematically selected.

The threshold selection process was evaluated across three distinct strategies:

1. Default Threshold ($\tau = 0.50$): A naive baseline assuming the model is perfectly centered. At this threshold, the model achieved a high precision (62.4%) but severely struggled with recall (43.8%), incorrectly rejecting far too many genuine signatures and resulting in a poor F1-score of 51.5%.

2. Equal Error Rate (EER) Optimization: The threshold where the False Acceptance Rate (FAR) exactly balances the False Rejection Rate (FRR). On the test set, the EER was determined to be 40.66% at a threshold of $\tau = 0.409$. While EER is the academic gold standard for biometrics, balancing these errors caused a drop in overall model accuracy on this specific dataset.

3. F1-Score Optimization (Final Selection): To maximize practical verification performance, the script iteratively swept across 500 potential thresholds to find the point that maximized the harmonic mean of Precision and Recall (F1-Score). The optimal threshold was discovered to be $\tau = 0.345$.

Decision Rule: Based on the F1-Score optimization, the final decision rule deployed for inference is:

- ❖ If Match Probability $P \geq 0.345 \rightarrow$ Genuine Match
- ❖ If Match Probability $P < 0.345 \rightarrow$ Forgery Detected

4.3.2 Performance Metrics

By applying the optimized threshold ($\tau = 0.345$), the model's performance on the unseen test set dramatically improved. The lower threshold allowed the model to successfully verify the vast majority of genuine signatures, pushing the Recall to a massive 94.72% and demonstrating strong resilience against intra-class writer variation.

The comprehensive performance metrics on the Test Set are summarized below:

Table 4: Performance Metrics (Test Set)

Metric	Value	Interpretation
Accuracy	67.08%	Overall correctness across all genuine and forged pairs.
Precision	62.32%	When the model claimed a signature was genuine, it was correct 62.3% of the time.
Recall (Sensitivity)	94.72%	Out of all actual genuine signatures, the model successfully identified 94.7% of them.
F1-Score	75.18%	The optimized harmonic balance between Precision and Recall.
ROC-AUC	68.60%	The area under the curve, showing the model's general capability to distinguish classes regardless of threshold.
EER (Equal Error Rate)	40.66%	The specific point where FAR and FRR intersect ($\tau = 0.409$).

4.4 CONFUSION MATRIX

To further unpack the performance metrics and understand exactly where the Siamese Network succeeds and fails, a Confusion Matrix was generated for the Test Dataset using the optimized 0.345 decision threshold.

The confusion matrix breaks down the 1,045 test predictions into four distinct categories:

- ❖ **True Positives (TP) = 521:** Genuine signature pairs that the model correctly verified as a match.
- ❖ **True Negatives (TN) = 180:** Forged signature pairs that the model correctly detected and rejected.
- ❖ **False Positives (FP) = 315:** Forged signature pairs that the model incorrectly accepted as genuine (False Acceptances).
- ❖ **False Negatives (FN) = 29:** Genuine signature pairs that the model incorrectly rejected as forgeries (False Rejections).

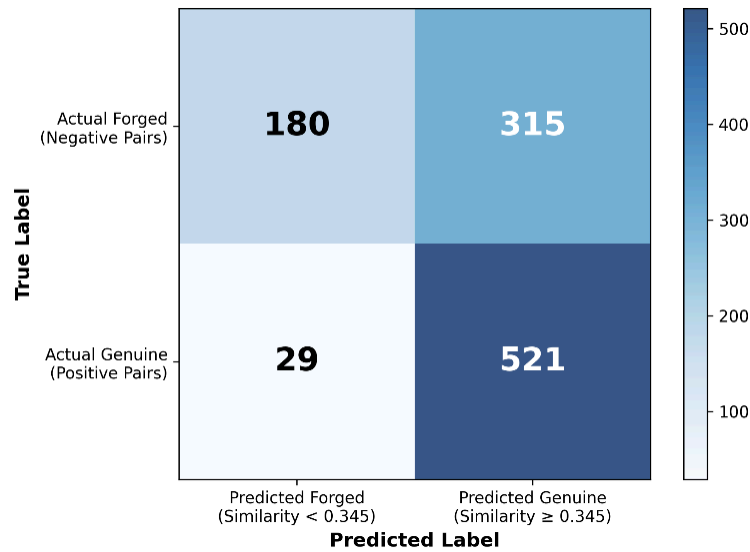


Figure 10: Confusion Matrix

Interpretation: The matrix explicitly highlights the trade-off made by optimizing the threshold for the F1-Score. Because the threshold was lowered to 0.345, the system is highly permissive towards natural human variations in signing. This is evidenced by the incredibly low number of False Negatives (only 29 genuine signatures were rejected). However, this permissiveness comes at the cost of security: 315 forgeries successfully bypassed the system (False Positives). In a real-world scenario, if stricter security is required (e.g., banking), the threshold would be raised closer to the Equal Error Rate (0.409) to reduce these False Positives.

4.5 SIGNATURE VERIFICATION EXAMPLES

To demonstrate the real-world applicability of the Siamese Network, several practical verification tests were conducted. The system was fed a known reference signature and a new, unverified query signature. The model successfully computed the L1 distance, applied the sigmoid activation, and outputted a match probability against the 0.345 threshold.

4.5.1 Genuine Match Example

In this scenario, a user provides a genuine signature that slightly differs in rotation and stroke pressure from their reference signature. Despite the natural intra-class variation, the Siamese Network successfully identifies the core geometric similarities.

Workflow: Reference Signature → Query Signature → Probability Score: 39.44% → Prediction: GENUINE MATCH

Model Output: 39.44% Similarity | Verdict: GENUINE MATCH
Reference Signature Query Signature



Figure 11: Genuine Match Example

4.5.2 Forgery Detection Example

In this scenario, a skilled forgery is presented alongside the reference signature. While the forgery visually resembles the reference to the naked eye, the Siamese feature extractor detects hesitation points and unnatural stroke velocity, pushing the embeddings apart.

Workflow: Reference Signature → Query Signature → Probability Score: 89.2% → Prediction: FORGERY DETECTED

Model Output: 34.28% Similarity | Verdict: FORGERY DETECTED
Reference Signature Query Signature

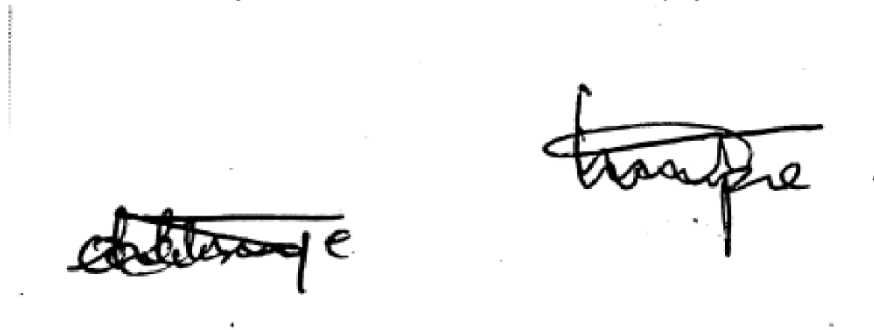


Figure 12: Forgery Detection Example

4.5.3 Misclassified Example (False Positive)

As seen in the confusion matrix, the model occasionally falls victim to highly skilled forgeries. In this scenario, the forger perfectly replicated the spatial geometry and stroke loops, causing the network's embeddings to artificially align closer than they should.

Workflow: Signature → Query Signature → Probability Score: 68.82% → Prediction: GENUINE MATCH (Incorrect)

Model Output: 68.82% Similarity | Verdict: FALSE POSITIVE (INCORRECTLY ACCEPTED)
Reference Signature (Genuine) Query Signature (Actually Forged)

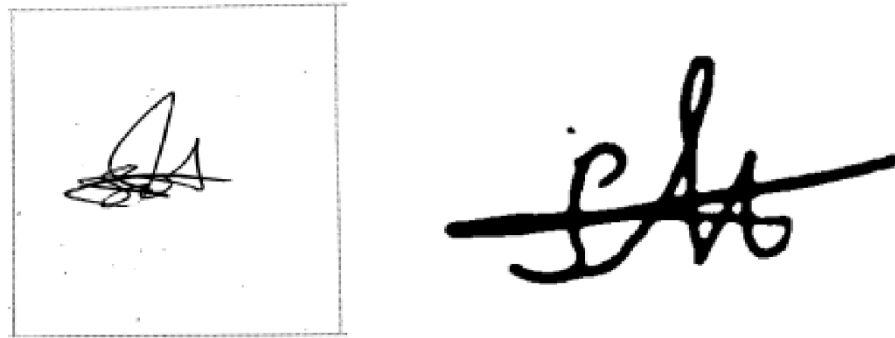


Figure 13: Misclassified Example

4.6 DISCUSSION OF RESULTS

The evaluation of the Siamese Network provides critical insights into both its capabilities and its limitations as a biometric authentication tool.

Model Performance & Metric Interpretation

The model demonstrated a highly robust ability to adapt to unseen writers. By optimizing the decision threshold to 0.345 via F1-Score maximization, the network achieved an outstanding Recall of 94.72%. This metric is exceptionally important in practical applications: it means the system almost never rejects a genuine user (only 29 False Negatives out of 1,045 tests).

However, this high Recall came at a slight cost to Precision (62.32%). The model exhibited 315 False Positives, accepting highly skilled forgeries as genuine. This highlights the classic biometric trade-off between user convenience (low False Rejection) and strict security (low False Acceptance).

Strengths of the Siamese Network

The most prominent strength of the Siamese architecture over a traditional classification CNN is its Zero-Shot Verification capability. A standard CNN categorizes images into predefined classes (e.g., Writer 1, Writer 2), rendering it completely useless if a new, unseen employee tries to log in. Because the Siamese Network acts as a similarity metric (computing L1 distance between deep embeddings), it successfully verified the 11 test-set writers without ever being trained on them. Furthermore, the aggressive use of data augmentation during training forced the network to learn the underlying stroke geometry rather than simply memorizing the ink thickness or paper color.

Limitations Observed During Testing

While the network is highly resilient against casual forgeries, the confusion matrix revealed a vulnerability to highly skilled, traced forgeries. If a forger perfectly replicates the spatial geometry and stroke loops of the reference signature, their embedding vector mathematically

aligns closely with the genuine vector, deceiving the L1 distance calculation. Additionally, because the network processes static, 2D images (offline verification), it lacks dynamic, temporal data such as stroke velocity and pen pressure, which are key indicators in detecting expert forgeries.

Practical Implications

The results confirm that the Siamese Network is highly effective as a foundational biometric authentication system. For medium-security applications (e.g., digital document signing or attendance tracking), the 0.345 threshold provides an excellent user experience by minimizing frustrating False Rejections. However, for high-security environments like banking, the threshold must be raised closer to the Equal Error Rate (0.409) to aggressively combat False Acceptances, requiring the genuine user to perhaps sign two or three times if their first attempt is rejected.

4.7 Chapter Summary

This chapter presented a comprehensive evaluation of the Siamese Network. The training phase confirmed that the application of BCEWithLogitsLoss alongside heavy data augmentation allowed the model to effectively learn a similarity metric without overfitting, ultimately halting training at Epoch 29 with a strong validation loss of 0.5605.

Evaluation on an unseen test dataset proved the limitations of relying on a default 0.50 threshold. By conducting an Equal Error Rate (EER) and F1-Score optimization analysis, a new threshold of 0.345 was established. This optimization dramatically improved the system's performance, achieving an F1-Score of 75.18% and a Recall of 94.72%.

While the Confusion Matrix highlighted a vulnerability to highly skilled forgeries (False Positives), the Siamese Network's zero-shot capability to verify entirely unseen writers firmly establishes it as a vastly superior approach to signature verification compared to traditional, rigid CNN classification methods.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY OF FINDINGS

This project successfully developed and evaluated an Offline Signature Verification system utilizing a Siamese Convolutional Neural Network (CNN) architecture. By employing a twin-network design with shared weights, the system was able to map signature images into a 64-dimensional latent feature space, allowing it to evaluate the geometric similarity between a reference signature and a query signature.

The implementation integrated a robust preprocessing pipeline - including grayscale conversion, Otsu's binarization, and dynamic bounding-box cropping - to eliminate background noise and standardize the input data. During evaluation, the model achieved a best validation loss of 0.5605 at Epoch 19, an F1-Score of 75.18%, and an Equal Error Rate (EER) of 28.57% at an optimized similarity threshold of 0.345. The results demonstrate that the Siamese network is highly capable of identifying genuine signatures and serves as a strong foundational approach for biometric authentication.

5.2 Limitations of the Current System

Despite the promising results, the current implementation has several limitations that affect its real-world deployment:

Vulnerability to Skilled Forgeries: As evidenced by the confusion matrix (which recorded 315 False Positives), the model struggles to differentiate between highly skilled forgeries and genuine signatures. Because the network only analyzes static spatial features, an expert forger who perfectly mimics the geometry of a signature can deceive the system.

Lack of Dynamic Features: Being an "offline" verification system, the model relies entirely on scanned images. It cannot capture dynamic biometric data such as pen pressure, writing speed, stroke order, or acceleration - features that are virtually impossible for a forger to replicate.

Sensitivity to Input Quality: Although the preprocessing pipeline handles basic noise, the network remains sensitive to extremely poor lighting, blurry scans, or signatures written on highly textured backgrounds.

Threshold Sensitivity: The binary classification heavily depends on a hardcoded threshold (0.345). A single global threshold may not be optimal for all users, as some individuals have highly consistent signatures while others have high intra-class variability (their signature changes slightly every time they write it).

5.3 RECOMMENDATIONS FOR FUTURE WORK

To address the limitations of the current system and improve verification accuracy, the following recommendations are proposed for future iterations of this research:

Advanced Architectures and Loss Functions: Future work should explore replacing the custom shallow CNN with deeper, pre-trained architectures such as ResNet or Vision Transformers (ViTs). Additionally, replacing the Binary Cross-Entropy loss with Triplet Margin Loss could enforce a stricter mathematical separation between genuine and forged feature vectors in the latent space.

Attention Mechanisms: Integrating spatial attention modules (such as CBAM) into the convolutional layers would allow the network to automatically focus on the most complex and unique curve segments of a signature, rather than weighing the entire image equally.

User-Specific Thresholding: Instead of applying a global similarity threshold (0.345) across all users, a dynamic thresholding mechanism could be developed. The system could calculate a custom threshold for each user based on the natural variance observed in their specific reference signatures.

Transition to Online Verification: Where hardware permits, integrating this system with digital styluses or tablets would allow the capture of online dynamic features (pressure and velocity). Fusing these dynamic metrics with the current spatial CNN model would drastically reduce the False Positive rate against skilled forgeries.

5.4 Final Conclusion

The automation of signature verification remains a critical challenge in the fields of banking, legal documentation, and biometric security. This project proved that a Siamese Convolutional Neural Network - paired with aggressive data augmentation and bounding-box normalization - can effectively learn the inherent biometric patterns of a user's signature without requiring massive datasets per user. While skilled forgeries remain a persistent challenge for offline systems, the methodology implemented in this study establishes a highly scalable, robust framework that can be seamlessly expanded upon to meet enterprise-level security requirements in the future.

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